MIL-F-51221A(EA)
1 October 1984
SUPERSEDING
MIL-F-51221(EA)
16 February 1984

MILITARY SPECIFICATION

FILTER, PARTICULATE, 150 CFM, M24

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

This specification covers one type of particulate filter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping, Fiberboard.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-TDC-S (A), Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance - Test Methods.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS AND ASSOCIATED LISTS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND
CHEMICAL RESEARCH AND DEVELOPMENT CENTER

TDPL 5-19-1226 - Filter, Particulate, 150 CFM, M24.

PUBLICATIONS

CHEMICAL RESEARCH AND DEVELOPMENT CENTER INSTRUCTION MANUAL

136-300-175 - Installation, Operation, and Maintenance of Penetrometer, Filter Testing, DOP, Q107.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing TDPL 5-19-1226 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the item shall conform to the specifications and drawings listed on TDPL 5-19-1226 and subsidiary drawings.

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3.2 Manufacture and assembly. The particulate filter shall be assembled as specified on Drawing 5-19-1226.

3.3 Airflow resistance. The airflow resistance of the filter shall be not more than 1.0 inch of water at a flowrate of 150 cubic feet per minute (cfm) when tested as specified in 4.4.4.1.

3.4 Dioctylphthalate (DOP) smoke penetration. The DOP smoke penetration shall be not more than 0.03 percent at the rated airflow when tested as specified in 4.4.4.1.

3.5 Resistance to rough handling. After rough handling as specified in 4.4.4.2 for not less than 15 minutes with the pleats, faces and separators in the vertical position, the filter shall comply with the airflow resistance and DOP penetration requirements of 3.3 and 3.4.

3.6 Preproduction. Prior to the start of regular production, a preproduction sample of particulate filters shall be produced in accordance with this specification for examination and tests (see 4.3).

3.7 Workmanship. The filter shall be free from foreign matter (dirt, oil, or viscous material) and damage such as dented, bent, burred, or distorted frames; deformation or sagging of medium or separators; cracks in the adhesive; cracks or holes in the exposed portions of the medium; and damaged or improperly secured gaskets.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

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4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of ten particulate filters shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. The preproduction sample shall be examined in accordance with the classification of defects (4.4.3.3).

4.3.2.2 For tests. The preproduction sample shall be tested in accordance with 4.4.4.

4.3.3 Acceptance/rejection criteria. Each filter in the preproduction sample shall comply with the requirements of this specification when examined and tested as specified in 4.3.2 to be acceptable.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the filters produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of filter material shall be represented in any one lot of finished filters.

4.4.2 Sampling for examination and tests. Sampling for examination and tests and level A packaging shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedure.

4.4.3.1 For examination and tests. Sample particulate filters shall be examined and tested in accordance with the classification of defects (4.4.3.3) and MIL-STD-105.

4.4.3.2 For critical defects. Each filter in the lot shall be tested for the critical defects listed in the classification of defects (4.4.3.3).

4.4.3.3 Classification of defects.

(a) Filter, particulate, 150 CFM, M24 (Dwg 5-19-1226).

Categories	Defects	Acceptance standards
Critical:		
1	Airflow resistance	4.4.4.1
2	DOP smoke penetration	4.4.4.1

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Major: AQL 0.65 percent defective

101	Resistance to rough handling	4.4.4.2

AQL 1.0 percent defective

102	Component missing
103	Component incorrectly assembled
104	Overall dimensions incorrect (Dwg 5-19-1227)
105	Marking incorrect
106	Workmanship (3.7)

Minor: AQL 4.0 percent defective

201	Marking illegible

(b) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Unit pack box incorrect
102	Quantity per package incorrect
103	Fiberboard separator missing, incorrect size, or in improper position
104	Closure incorrect
105	Packing box incorrect
106	Quantity of packages per pack incorrect
107	Packing closure and strapping incorrect
108	Marking missing or illegible

4.4.4 Tests.

4.4.4.1 Airflow resistance and DOP smoke penetration. The filters shall be tested for airflow resistance and DOP smoke penetration with the Q107 Penetrometer in accordance with 136-300-175 or equivalent test machine and method.

4.4.4.2 Resistance to rough handling. The filters shall be rough handled in accordance with MIL-STD-282, Method T105.10 and then tested for airflow resistance and DOP smoke penetration in accordance with 4.4.4.1.

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5. PACKAGING

5.1 Unit packing, level A. Two filters shall be packaged together in a style FOL, type CF, class weather-resistant, grade W5c fiberboard box conforming to PPP-B-636, having inside dimensions of 22 by 4-1/8 by 21 inches, length, width and depth, respectively. A 22 by 21 inch size fiberboard separator fabricated from the same material as the box shall be placed between the two filters. The box shall be closed in accordance with the requirements for FOL style boxes in the fiberboard box specification.

5.2 Packing. Packing shall be level A or B as specified (see 6.2).

5.2.1 Level A. Five (5) boxes of filters unit packed as specified in 5.1, shall be packed in a class 2, style 2 or 2-1/2 nailed wood box conforming to PPP-B-621. The box shall have the approximate inside dimensions of 22-1/2 by 22 by 22-3/4 and shall be closed and strapped in accordance with requirements in the box specification. The strapping shall be galvanized.

5.2.2 Level B. Filters shall be packed as specified in 5.2.1, except that class 1, style 2 or 2-1/2 nailed wood boxes and coated finish strapping shall be used.

5.3 Marking. In addition to any special marking required by the contract or order, unit packs (5.1), and packs (5.2) shall be marked in accordance with MIL-STD-129.

5.4 Repair parts. When this item is being procured for stockage and issue as a repair part, preservation, unit packing, packing and marking shall be as specified on the packaging data sheet which is identified by its National Stock Number/Packaging Sheet (PS) number.

6. NOTES

6.1 Intended use. The filter covered by this specification is used for the mechanical filtration of particulate matter from atmospheric air. This particulate filter is a component of the M18, electric motor driven, 300 CFM, gas-particulate filter unit for the Hawk Battery Control Central and the M51 (POD) Collective Protection Shelter System.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level packing required.
(c) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instruction when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

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6.3 Packaging data sheet. The packaging data sheet is identified by the Federal Stock Number for this item and should be ordered from Commander, Chemical Research and Development Center, ATTN: SMCCR-SPD-P, Aberdeen Proving Ground, MD 21010-5423.

6.4 Manufacture. The following method of manufacture has been found to produce a satisfactory particulate filter. This procedure is furnished with the distinct understanding:

(a) That the contractor is not required to follow the method.
(b) That the provisions of the specification must be met regardless of the method of manufacture.
(c) That the Government makes no warranty that material made in accordance with this method will pass the specification requirements.

Manufacturing procedure:

(a) Filter material conforming to MIL-F-51079 should be pleated on a suitable filter pleating machine using fiberboard separators to form and hold the pleats. A section of pleated material containing the number of folds which have been found to give the desired compactness should be cut-off. This section should be placed on a temporary U frame approximately 19 inches in length, 20 inches wide with ends 1-3/4 inches high (inside dimensions). The exposed fiberboard separators should be removed and fluted aluminum separators which are approximately 21-1/2 inches long inserted in their place. Any torn places in the filter material should be repaired with adhesive for filters. The pleated assembly shall be turned over and the operation repeated.

(b) The pleated section should be checked for proper compactness. Excessively tight packing will cause the fluted separators to cut into the filter material, and excessively loose packing will allow undesirable movement of the separators. Correct compactness may be obtained by removing or splicing in sections of pleated filter material to the ends of the assembly.

(c) Fiberboard backing strips should be inserted between each end of the pleated assembly and the temporary frame to keep pleats from bending over when trimmed. Both sides of pleated assembly should be trimmed with a high speed band saw to obtain correct width. It is necessary to have an angle of no less than 30° between the saw blade and filter pleats when trimming to prevent the saw from tearing the paper.

(d) One coat of adhesive should be applied to both sides of the pleated assembly. Allow to penetrate and dry approximately 2 hours. A second coat of adhesive should be applied. This coat should be allowed to penetrate and dry approximately 4 hours. A third coat of adhesive should be applied and allowed to penetrate and dry approximately 4 hours. Additional coats should be applied if necessary to cover any pin holes which might arise when solvent evaporates. After careful inspection to see that there are no pin holes remaining, the excess adhesive should be trimmed from the edges of assembly and the temporary frame removed.

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(e) The pleated assembly should be assembled in the frame and one coat of adhesive applied around the entire periphery sealing the frame to the filter. The adhesive should be allowed to dry two hours and a second coat applied and allowed to dry 4 hours. This operation should then be repeated on the reverse side.

(f) The final test for DOP penetration and air resistance shall be performed and the filter marked.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A809

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-51221A(EA)	FILTER, PARTICULATE, 150 CFM, M24

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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